Armando Montes de Oca Medina

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Mechanical Engineering student (Mexican–German DHIK dual degree, Tec de Monterrey + TH Brandenburg) with 12 months of industry design experience at Schneider Electric. Seeking a **mandatory internship (Pflichtpraktikum) of at least 13 weeks, March–July 2027** (flexible from February to August). Student residence permit with work authorization in Germany.

EDUCATION

Technische Hochschule Brandenburg, Brandenburg an der Havel, Germany 09/2026 – present
B.Sc. Mechanical Engineering, DHIK dual-degree program

Tecnológico de Monterrey, Campus Monterrey, Mexico 08/2023 – 12/2027 (expected)
B.Sc. Mechanical Engineering | GPA 3.9/4.0 (97.49/100)

- Recognized twice for a top-1% GPA in the School of Engineering and Sciences (Aug–Dec 2024; Feb–Jun 2025).

Prepa Tec, Campus Cumbres, Monterrey, Mexico 08/2020 – 06/2023
International Baccalaureate (IB): 40/45 | Highest GPA of the graduating class (4.0) | German as third language, exchange in Berlin

EXPERIENCE

Schneider Electric, Monterrey, Mexico 08/2025 – 08/2026
Electromechanical Design Intern, North American Lighting Panelboards (Home & Distribution division)

- Updated 1,000+ bills of materials (BOMs) in PDM to consolidate separate warning and danger markings into one unified label across the panelboard product family, simplifying manufacturing and compliance review.
- Executed and validated 30–40 engineering changes on components against revised design requirements.
- Designed and 3D-modeled components and assemblies for low-voltage power distribution equipment in Creo Parametric.

INDUSTRY PROJECTS (TEAM PROJECTS, TEC DE MONTERREY)

FRISA Aerospace – Root-cause analysis of a ring-rolling mandrel failure 05/2026 – 06/2026

- Investigated why a 4340 steel mandrel failed after 30 forgings when FEA predicted 49,500 cycles, combining FEA fatigue-life estimation, fracture mechanics, hardness profiling, metallography and SEM fractography.
- Identified pre-existing surface defects as the main driver; recommended ultrasonic crack inspection, laser-ultrasonic assisted ball burnishing (LUAB) and 5S-based life tracking.

Daikin – Coupling failure analysis and redesign; tube-bank heat exchanger CFD 01/2026 – 04/2026

- Used static and modal FEA (Ansys) and a 5-whys analysis to trace recurring coupling failures to resonance; redesign with structural ABS foam raised the first natural frequency from 12.6–46.8 Hz to 118 Hz in simulation, keeping the assembly under 55 kg.
- Compared heat transfer and pressure drop for three turbulator configurations in a cross-flow heat exchanger (Ansys Fluent, k-ω SST, up to 9.5 M elements).

Carrier – CFD model of a psychrometric test chamber 11/2025 – 12/2025

- Modeled airflow for 5 TR and 25 TR HVAC units (Ansys Fluent), located recirculation zones and proposed a filleted nozzle inlet that reduced peak velocity by 3.7% in simulation.

Ternium – Sheet-steel formability simulation 02/2025 – 06/2025

- Built an explicit FEM model of the Nakajima test (LS-DYNA) for 7 specimen geometries to predict a forming-limit curve per ISO 12004-2, reducing the need for physical testing.

SKILLS, CERTIFICATIONS AND LANGUAGES

CAD	Creo Parametric, SolidWorks
Simulation	Ansys Mechanical (static, modal, fatigue), Ansys Fluent (CFD), LS-DYNA
PLM and data	Windchill, PDM, BOM and engineering change management; Excel (advanced), Power BI, Minitab
Methods	Root-cause and failure analysis, Lean Six Sigma, project management
Certifications	PMI CAPM (2026); Lean Six Sigma Yellow Belt (2026); Dassault Systèmes CSWA (2025); Microsoft Excel Data Analysis (2025); AWS Welding Inspection Assistant (2024); Solenum Solar Energy Diploma (2024)
Languages	Spanish (native); English (C2, TOEFL iBT 120/120); German (B2–C1; Goethe-Zertifikat B2); French (basic)